IBM Garage Digital Transformation

Smart Park Management
Technology Park Digital Middle-Platform Management System

#Label Management #Statistic Analysis #Edge Computing #Design Thinking Methodology

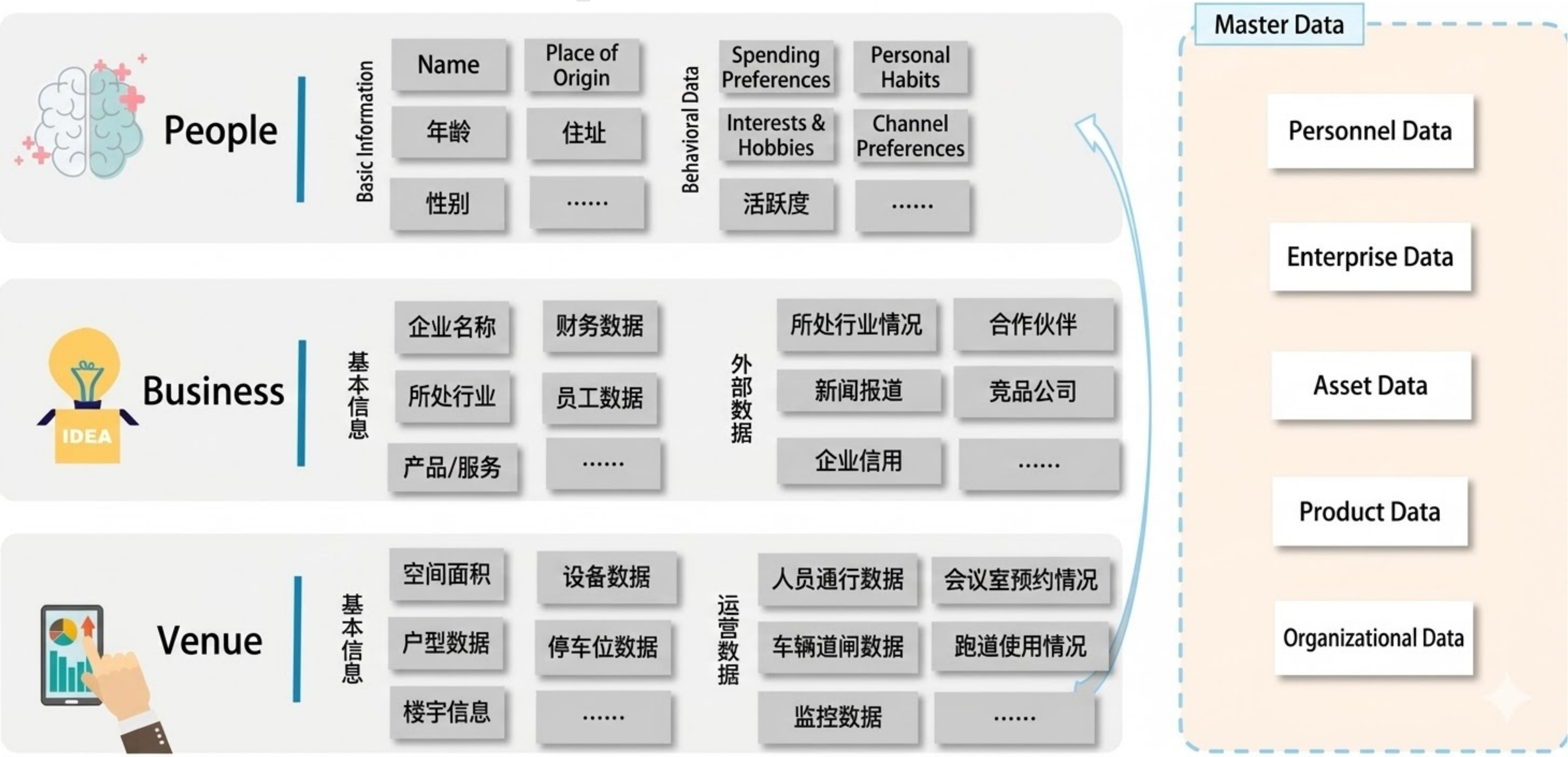
Project Intro

Project Overview

Beijing Dongsheng Bozhun, the operator of Dongsheng Software Park, aims to build and refine its existing **operational mechanisms** through three dimensions: **People, Industry, and Space**. At the same time, the company seeks to leverage **digital transformation** to achieve more profound management of **incubators**, **park enterprises**, and **park employees** within the smart campus ecosystem.

Current Data Landscape

The current data landscape is similarly categorized into three dimensions: **People, Industry, and Space**. By leveraging the **IBM Garage design methodology**, the goal is to help the enterprise achieve **data fusion**, creating innovative **operational models** and **management strategies** for the smart campus.



Current Data Landscape of Dongsheng Bozhun

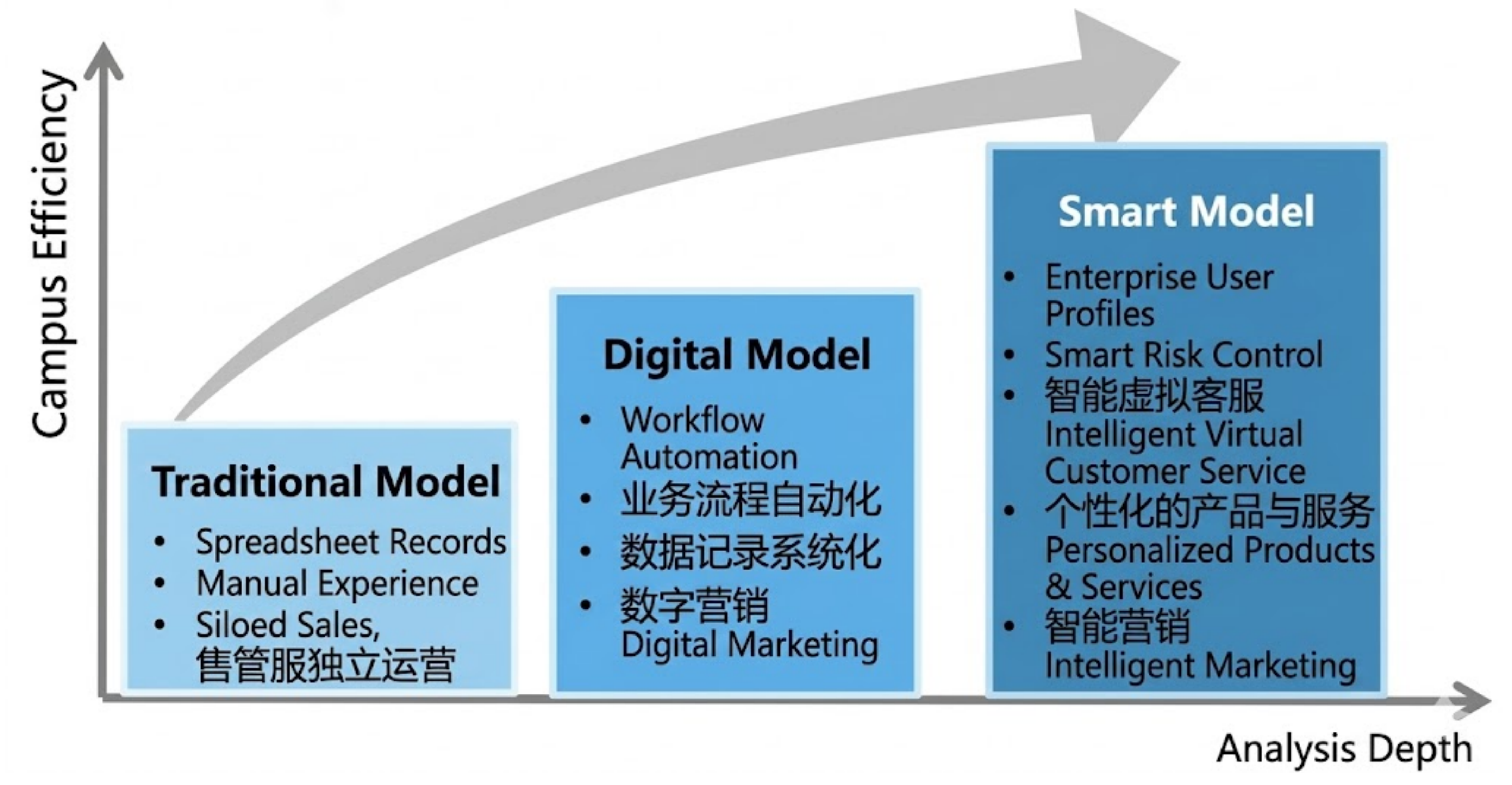
My Role

1.Framing Workshop
2.Design Thinking Workshop
3.UX/UI Design - Label Decision Dashboard

Project Intro

Digital Operational Transformation Vision

- **Data Acquisition:** Collect and clean heterogeneously distributed data. The breadth of data determines the depth of User Profiling dimensions and the extent of application scenarios.
- **Data Processing:** Establish data models to mine and organize critical data elements, ensuring a continuous cycle of optimization and iterative improvement.
- **Persona Creation:** Define distinct dimensions and utilize Knowledge Graphs to apply intelligent tags to enterprises.
- **Continuous Updates:** Ensure market and enterprise data are updated in real-time to maintain the vitality and relevance of enterprise profiles.

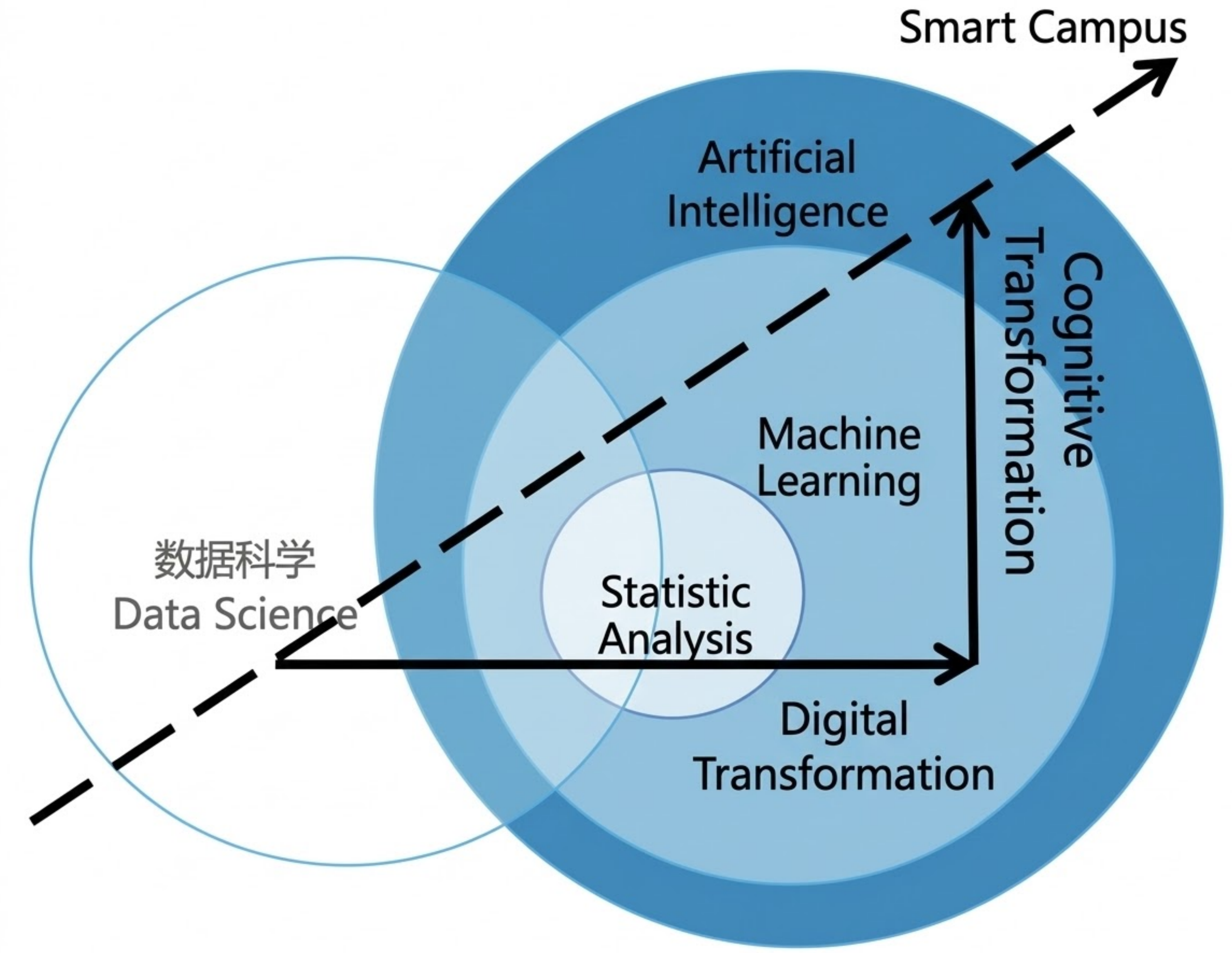


Digital Operations Roadmap

Business Data Sorting

Enterprise Knowledge Graph Construction

By constructing an **Enterprise Knowledge Graph**, we perform deep mining of complex network relationships between enterprises, talent, finances, products, and industrial chains. This enables **investment promotion and attraction** services for the science park, guiding industrial development within the ecosystem. For financial service departments, the graph monitors the growth trends of target enterprises and evaluates their **risk-bearing capacity**



Data Analytics-Driven Enterprise Profiling

Framing Workshop - Preliminary Departmental Interviews

Prior to the discovery phase, given the broad and complex scope of operations, it is essential to clarify the departmental attributes, business scopes, and preliminary requirements across the three dimensions of **People, Industry, and Space**. Consequently, a three-day interview series has been scheduled, primarily targeting **middle management** and **front-line personnel** at the departmental level.

PAIN POINTS	DETAILED DESCRIPTIONS
Information Asymmetry	A lack of enterprise info authenticity creates asymmetry between enterprises and the park management. info asymmetry.
Difficult Data Collection	Difficulty gathering data on personnel and venue, and finances means relying only on manual and time-consuming industry reports and field research.
Inefficient Business Tracking	Unclear business handovers, with many oral agreements, lead to tracking difficulties.
Insufficient Digital Management	Records are kept using paper and basic spreadsheets, hindering digital integration and sharing.
Lack of Workflow Standardization	Business processes and templates lack standardized and uniform management.

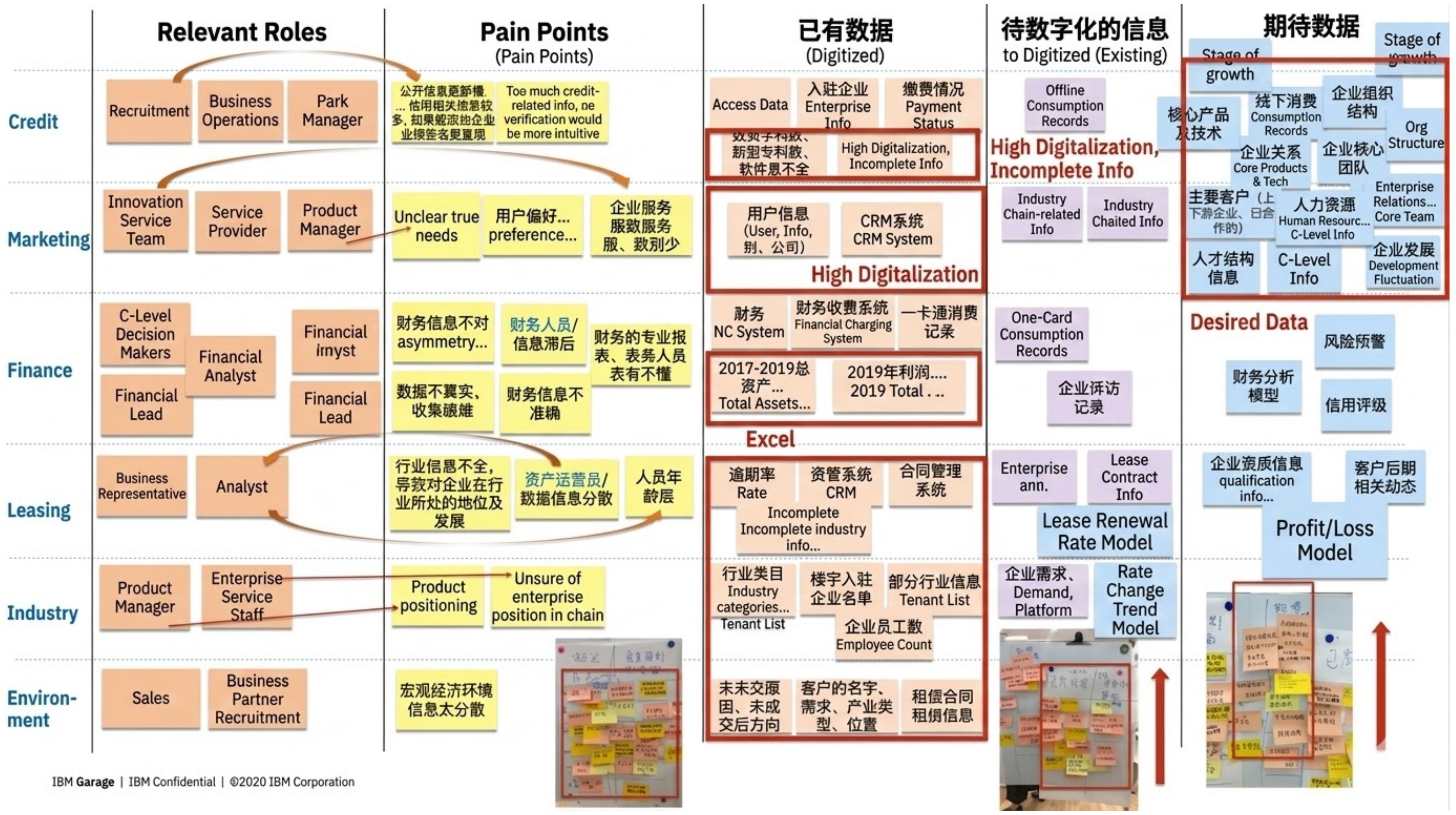
Summary of Interview Pain Points and Requirements

Framing Workshop

Framing Workshop - Business Opportunity Exploration

Following the interviews, it was concluded that beyond the operational difficulties caused by **information asymmetry**, departments aim to utilize this workshop to map out the information they possess alongside the root causes of their **pain points**.

Therefore, the entire discovery process will focus specifically on identifying pain points and requirements related to **information and data.**

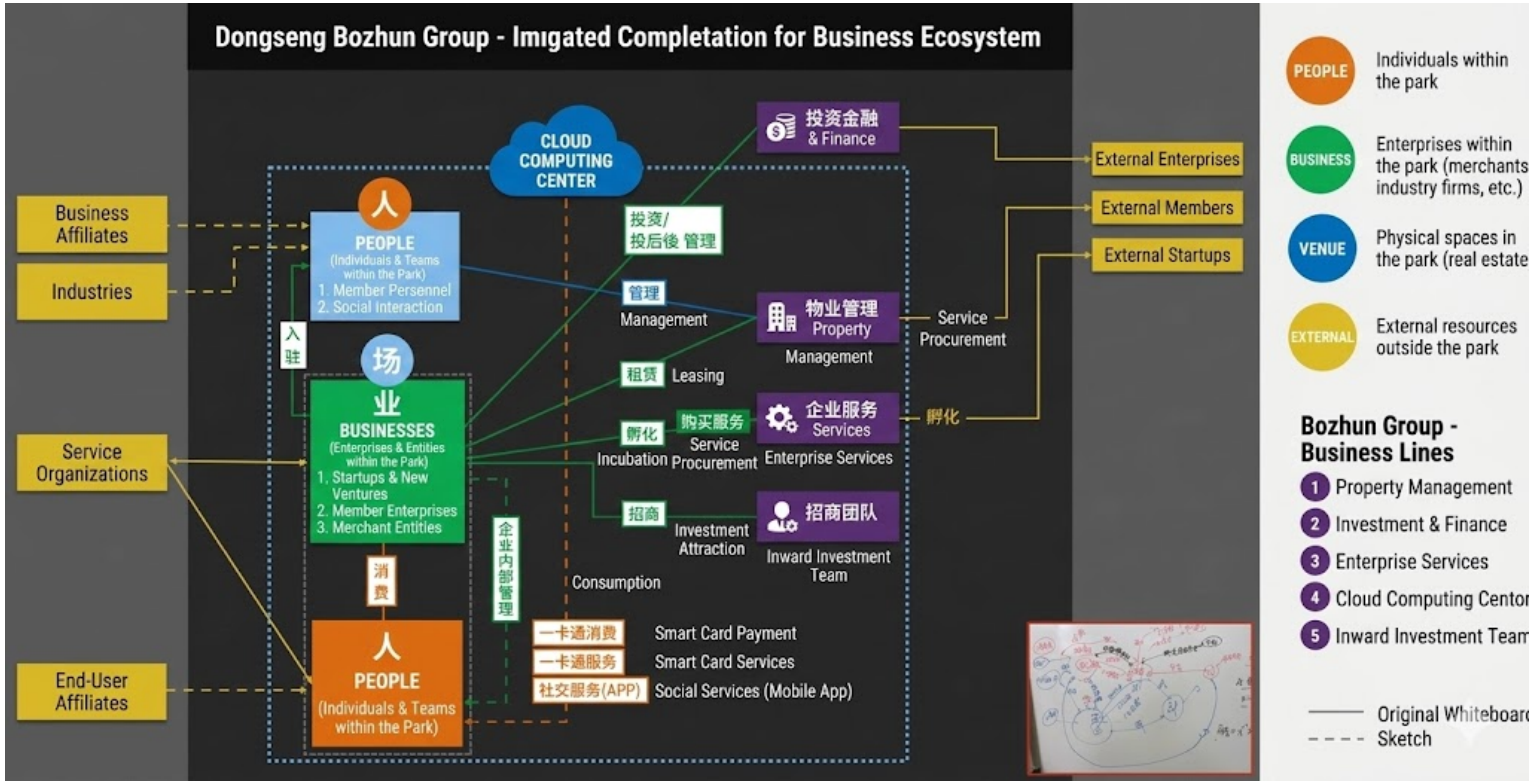


Business Opportunity Discovery

Framing Workshop

Framing Workshop - Stakeholder

Beyond addressing the **pain points** from the preliminary research and cleansing data across departments, we have mapped out the internal and external **relationship network** across the three dimensions of **People, Industry, and Space**. This identifies the specific relationships that must be integrated into the future **Operational Decision-Making Platform** to ensure clearer and more precise **data integration** in subsequent phases.



Dongsheng Stakeholder Map

Framing Workshop

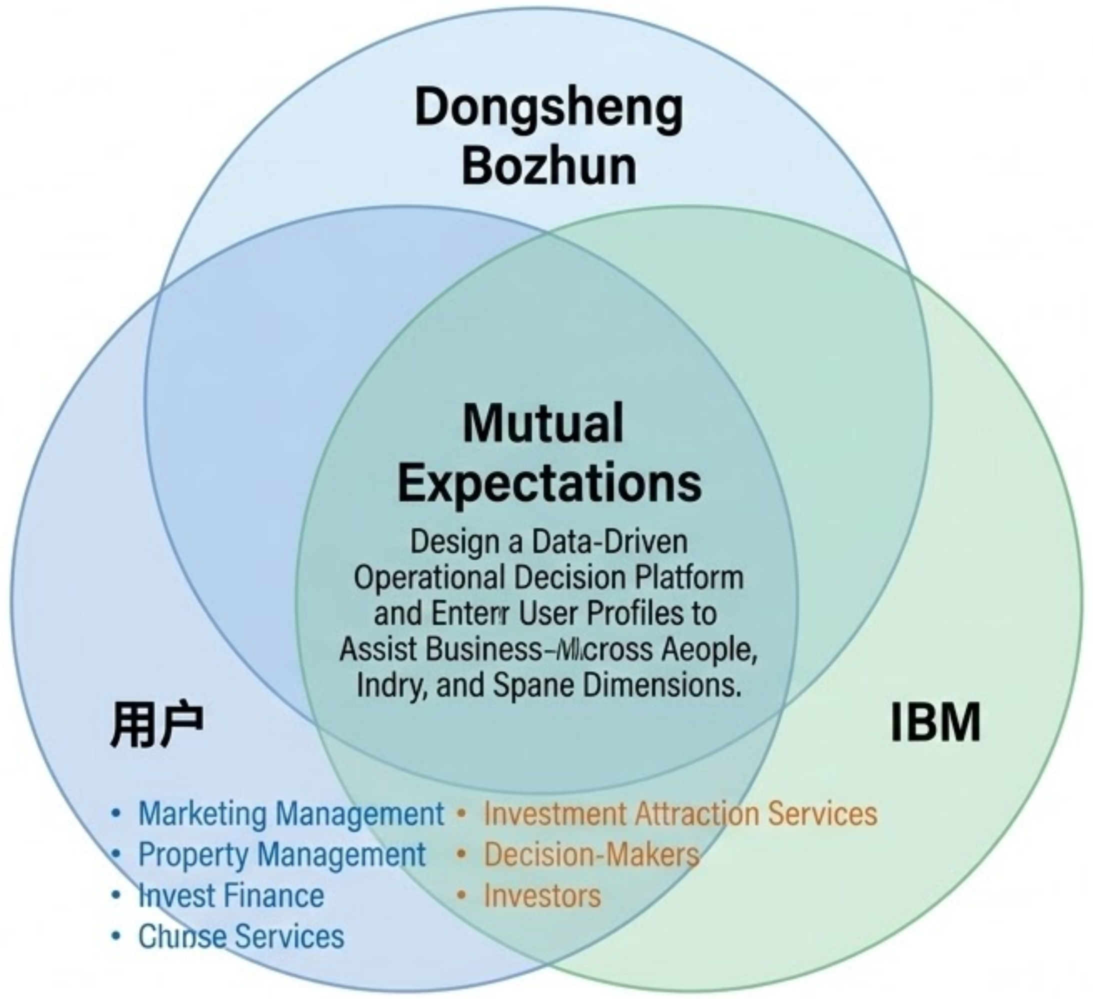
Framing Workshop - Opportunity Statement

Strengths (Prioritized by Importance + Feasibility)

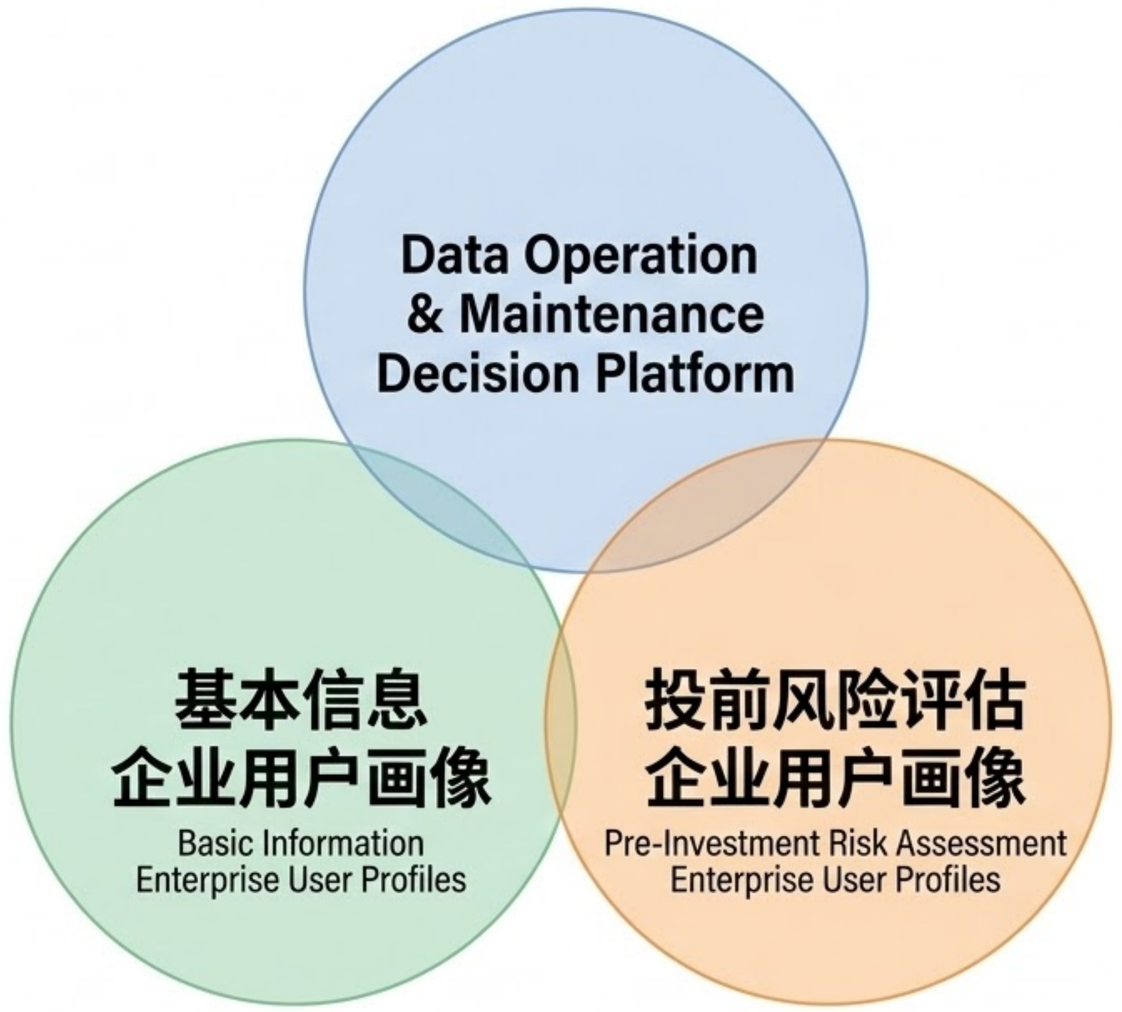
1. **Pre-investment Risk Control:** Enhanced risk mitigation for investment projects.
2. **Precise Market Positioning:** Accurately identifying target markets and tailoring product portfolios based on enterprise preferences (Science & Innovation Services).
3. **Enterprise Categorization:** Delivering personalized foundational services to boost tenant engagement and park loyalty

Weaknesses & Challenges

1. Information Gaps: Incomplete data leading to a lack of clarity regarding actual customer needs and preferences.
2. Data Silos: Highly fragmented data that is difficult to collect and impossible to share across departments.
3. Information Authenticity: Difficulty in verifying the accuracy of enterprise information, resulting in low screening efficiency.
4. Financial Information Asymmetry: Inaccurate financial data leading to misjudgment of enterprise risks.
5. Lack of Visualization: Insufficient reporting dashboards, making it difficult for business personnel to interpret data.
6. Industrial Chain Positioning: Lack of visibility into where an enterprise sits within the broader industrial value chain.
7. Tooling Deficit: Absence of professional tools for data acquisition, management, and utilization.
8. Real-time Synchronization: Challenges in keeping the system updated with the latest modifications to enterprise information.



Business Direction



Human resources, workplace actual execution business

Design Thinking Workshop

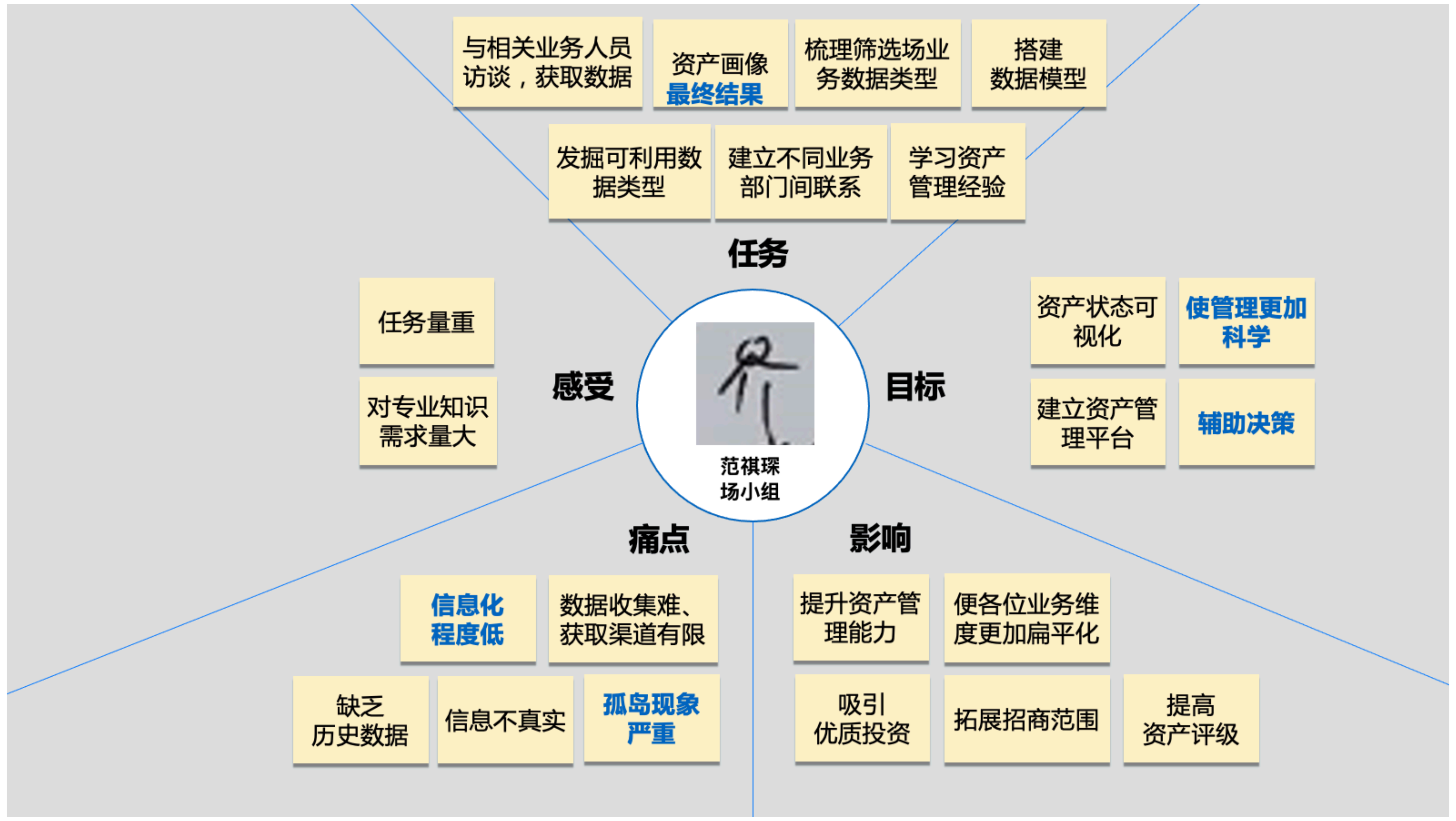
Design Thinking Workshop

Based on the insights from the **Business Opportunity Discovery** phase, four **Design Thinking Workshops** will be convened, involving departments such as **Digital Platform Operations, Finance & Investment, Property Leasing & Investment, and Enterprise Services**. The objective is to gain a deep understanding of **user psychology**, **workflow pain points and requirements**, and **system/data inventory**. The final output will define the essential **Enterprise User Profile** content and visualizations required for the **Data Operation & Maintenance Platform** (using Digital Platform Operations as a primary case study).

Design Thinking Workshop - Empathy Map

Asset Data Operations (Space/Venue Group) Pain Points:

1. **Lack of Data Authenticity:** Information is unreliable or unverified.
2. **Difficult Data Collection:** Limited acquisition channels and significant gathering hurdles.
3. **Low Level of Informatization:** A critical lack of historical data and digital infrastructure.



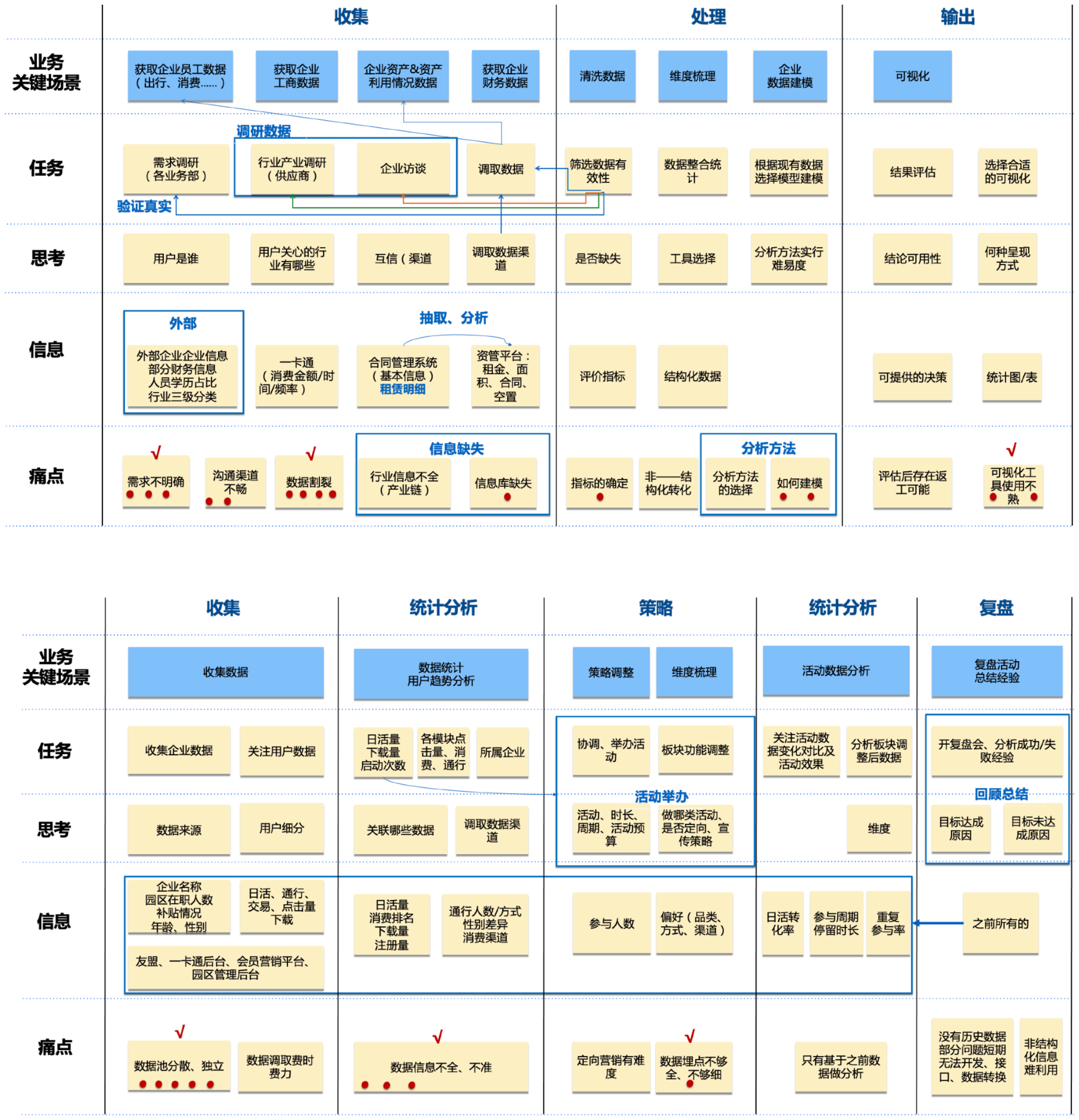
Empathy Map - Digital Platform Operations

Design Thinking Workshop

Design Thinking Workshop - User Journey Map

Key Pain Points Identified:

- **Information Gaps:** Critical data missing across key touchpoints.
- **Lack of Analytical Methodologies:** Deficiency in structured data analysis approaches.
- **Verification Challenges:** Data lacks authenticity and is difficult to validate.



User Journey Map - Digital Platform Operations

Design Thinking Workshop

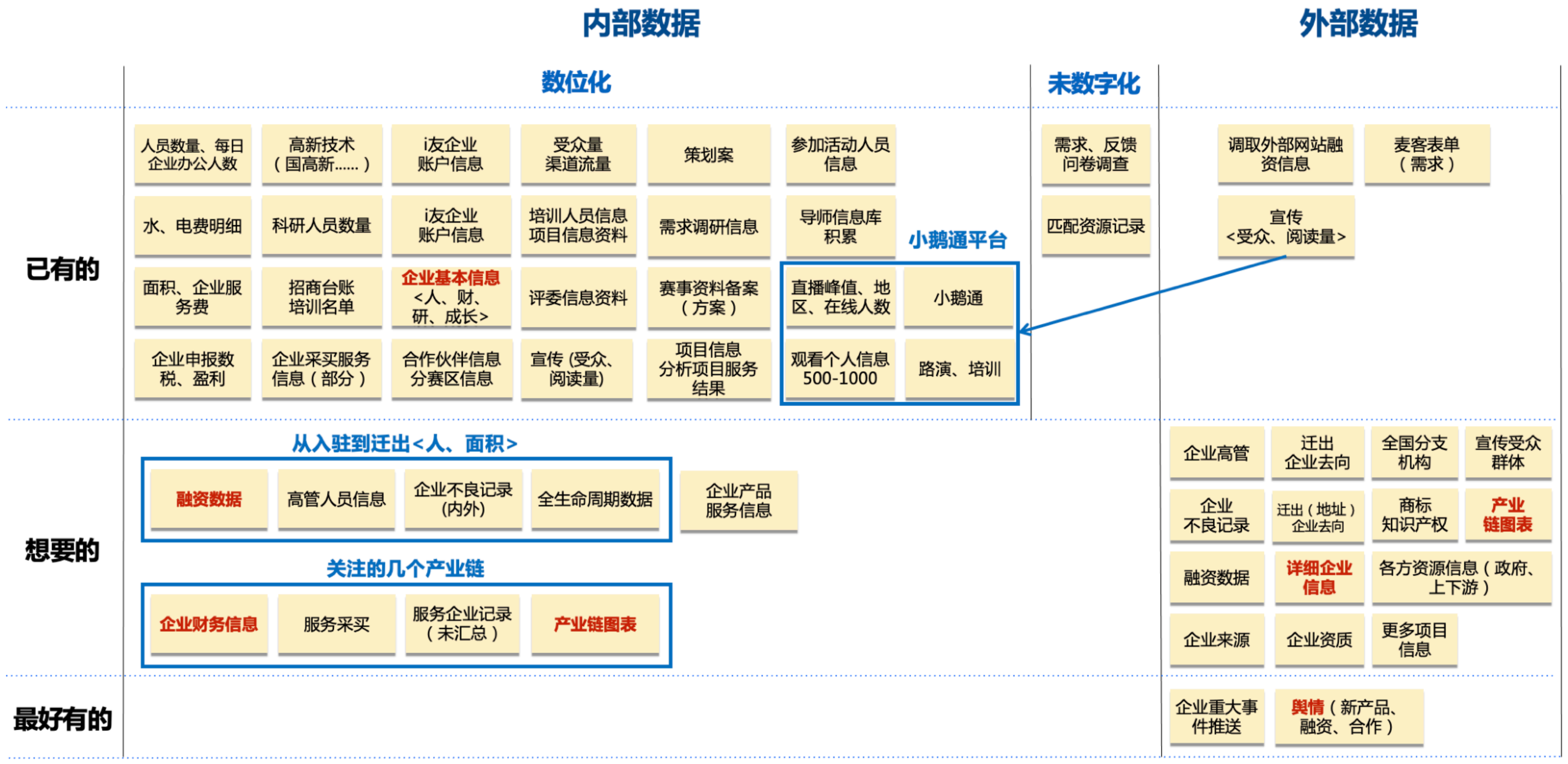
Design Thinking Workshop - User Journey Map

By comparing the pre-and post-workshop results from the Business Opportunity Discovery, and further validating pain points through Journey Mapping with a voting session, the department identified Difficulty in Data Collection and Data Gaps as the two most critical challenges. In the final stage of the Design Thinking Workshop, the specific missing data points will be meticulously identified and cataloged.

Design Thinking Workshop - Data & System Inventory

A comprehensive summary across the four departments reveals that the most critical missing datasets are **Corporate Financial Data** and **Employee Information**. The absence of these two key data categories significantly hinders **Dongsheng Bozhun** from performing effective **User Preference Analysis** and constructing accurate **Enterprise User Profiles.**

User Journey Map for Digital Platform Operations - Summary & Pain Point Voting

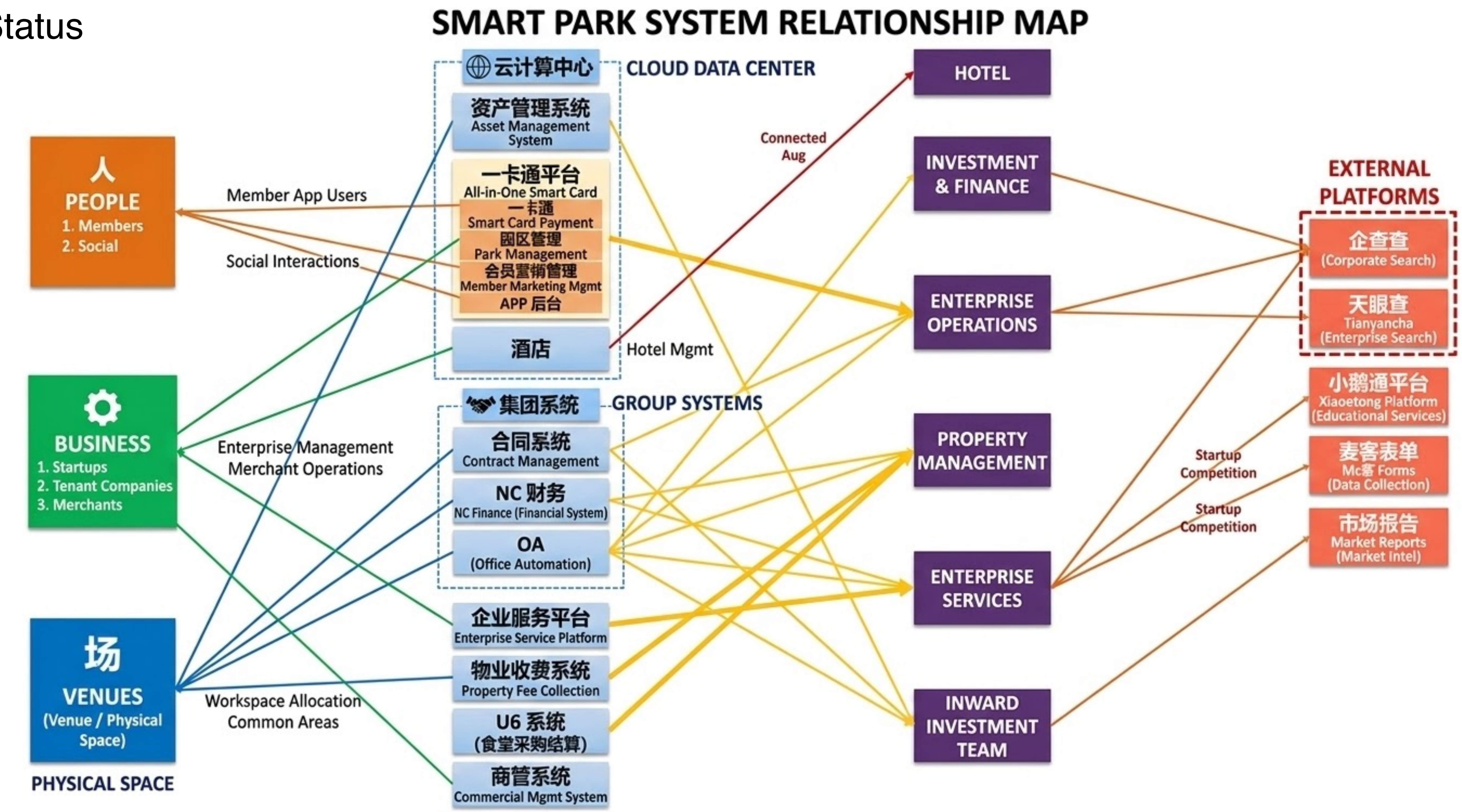


User Operations Data & System Inventory

Design Thinking Workshop - Digital Operational Decision-Making Platform

Summary of System Relationship Status

Based on the three foundational dimensions of **People, Industry, and Space**, we have mapped the relationships between systems and departments. This framework will assist **Data Scientists** in streamlining **Data Exploration**, data organization, and resolving **API/Data Call** issues during future program development.



System Relationship Status Map

Design Thinking Workshop - Digital Operational Decision-Making Platform

Current Information Status

Similarly, we have categorized existing information and desired future data integration across the dimensions of People, Industry, and Space. Our observations indicate that while the Industry (Enterprise) category is data-rich, it suffers from high redundancy and a lack of authenticity. This implies that Dongsheng Bozhun must prioritize Data Cleansing during future integration to generate accurate and actionable Enterprise User Profiles.

Furthermore, the key information prioritized by all departments includes: Corporate Financial Data, Industrial Chain Positioning, Enterprise Change Records, Public Sentiment/Media Monitoring, and Basic Personnel Information.

智慧校园数据要求矩阵

	Existing 数据（Eredjwned)	Desired 数据（needed)	理想 数据（Best-to-have)
People 1. Members 2. Socitactions	1. 人工及用户基本信息 2. 会员本信息、订单数据、消费记录 3. 企业相关福利福利 4. App 活动（日常/新量/,共用户、上出频率、保留率、活动用户）	1. 准确的企业员工基础数据★	1. 职工动态变化 2. 社会联系（友友、附属公司） 3. 员工购买历史、消费倩好、喜欢兴趣 4. 活动推广对象观群
Business 1. 创业业 2. 会员企业 3. 商户	1. 居民企业数据，项目基础数据，金融/运营数据★ 2. 项目空间数据，租有性区域、空空状况 3. 政策聚合、损弃/更要/通知、施施出出记录、企业商户、入门通券（人人员）、变更信息 4. 企业产商局局登册变更，股权结构，诉毒记录★ 5. 企业访问记录、投资条敂、租理模板、历史协议、保密性协议、投资意意言、股权转换协议、服务费数据 6. 外部社会媒体内容，小通直播数据，形式投签（外源） 7. 企业服务请求、反馈、问答、资源配置记录（文本式） 8. 创业大赛评会资料、观众数据、参与者信息、活件资料、导师商户、培训人信息 9. 科科人员数量，高科技企业证券	1. 工业价值链数据，行业分布、行业排行，上游下流关系★ 2. 企业基金流和债务状况，金融宣告，报告★ 3. 企业核心（管理/运营）人人資名 4. 具体园内内获得受的福利福利 5. 企业融资历史、服务采购和销售记录 6. 企业产品/服务详情、负债记录、高上平人员、进一步融资数据	1. 竞争者产品/市场智能 2. 其他公园企业的活动和数据 3. 商标与知识产权数料 4. 进一步企业证书、企业来源数据，详细融资资料，主要事件通知（新品/合作），媒体倩情分析★
Space Physical Assets & Venues	1. 租理合同、注册企业文案（文本基础） 2. 租居信息：租租面积，单价（租租、物业费）、空退率、成长率 3. 金融状况（租租、电源收款/优著） 4. 关键帐户损失/保留数料 5. 长期客户户案（Excel基础） 6. 高利用客户案（Excel基础）	1. 公共施设使用、维修及折扣数料 2. 在上场场预约数据 3. 智慧工厂功能整合（如、状态、使用） 4. 物理人员任务管理及分配 5. 企业能源消耗与支款状况（实时） 6. 权威第三方物理调查报告 7. 综合在上场场预约管理 8. 高级智慧工厂特能 9. 智慧物理人员任务分配	1. 其他公园站比平均单位租租（历史数据） 2. 智慧安全/防制系铳整合 3. 公园绿绿及维护监控

信息丰富，但都有亖多文性高，缺乏性不足

关键共享信息情優先谁：

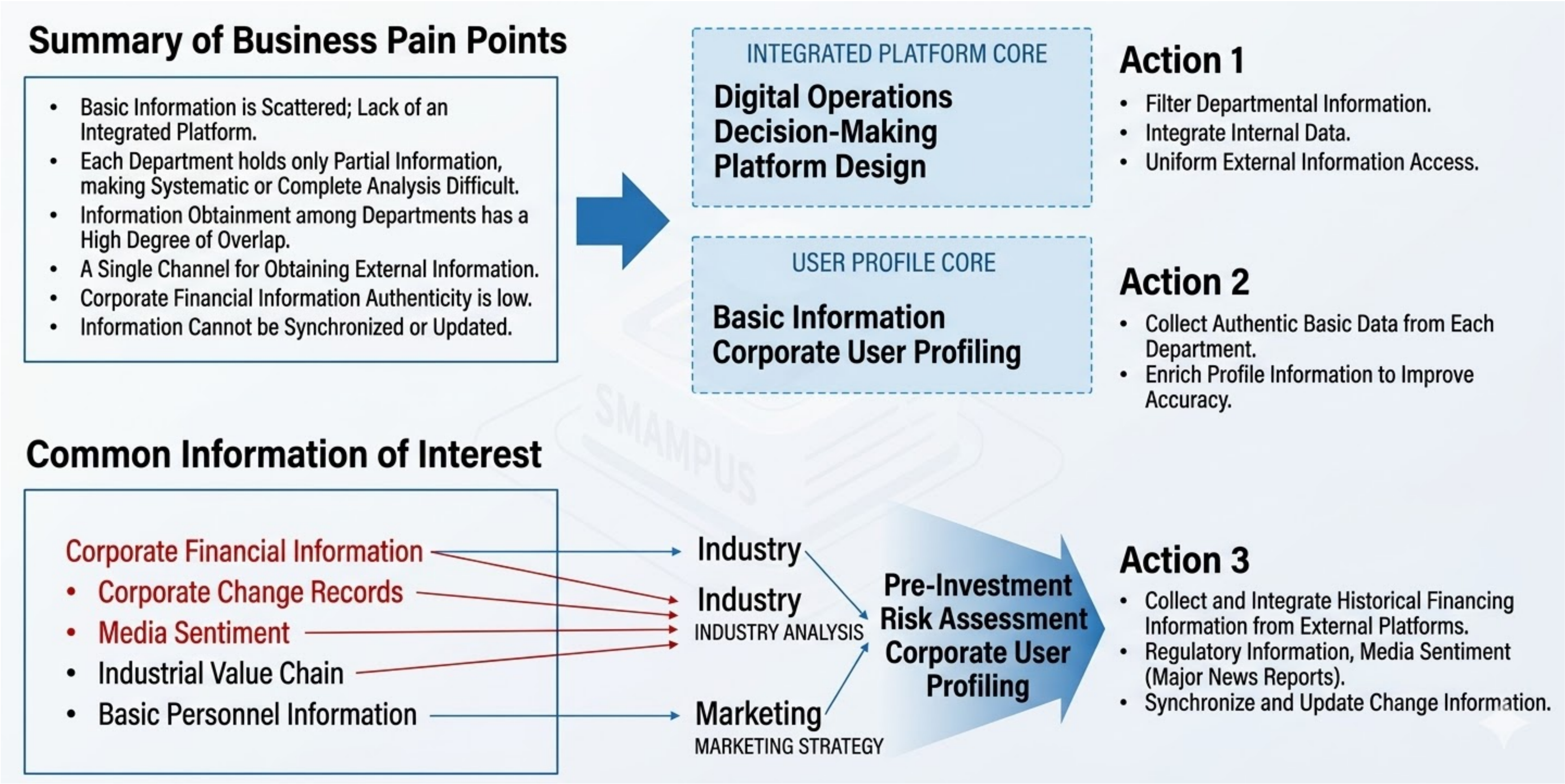
企业金融数料、工业价值链定位企业改变通知、企业公民信情、员工基础数据

Information Status Map

Design Thinking Workshop - Data Sorting

Data Inventory Summary

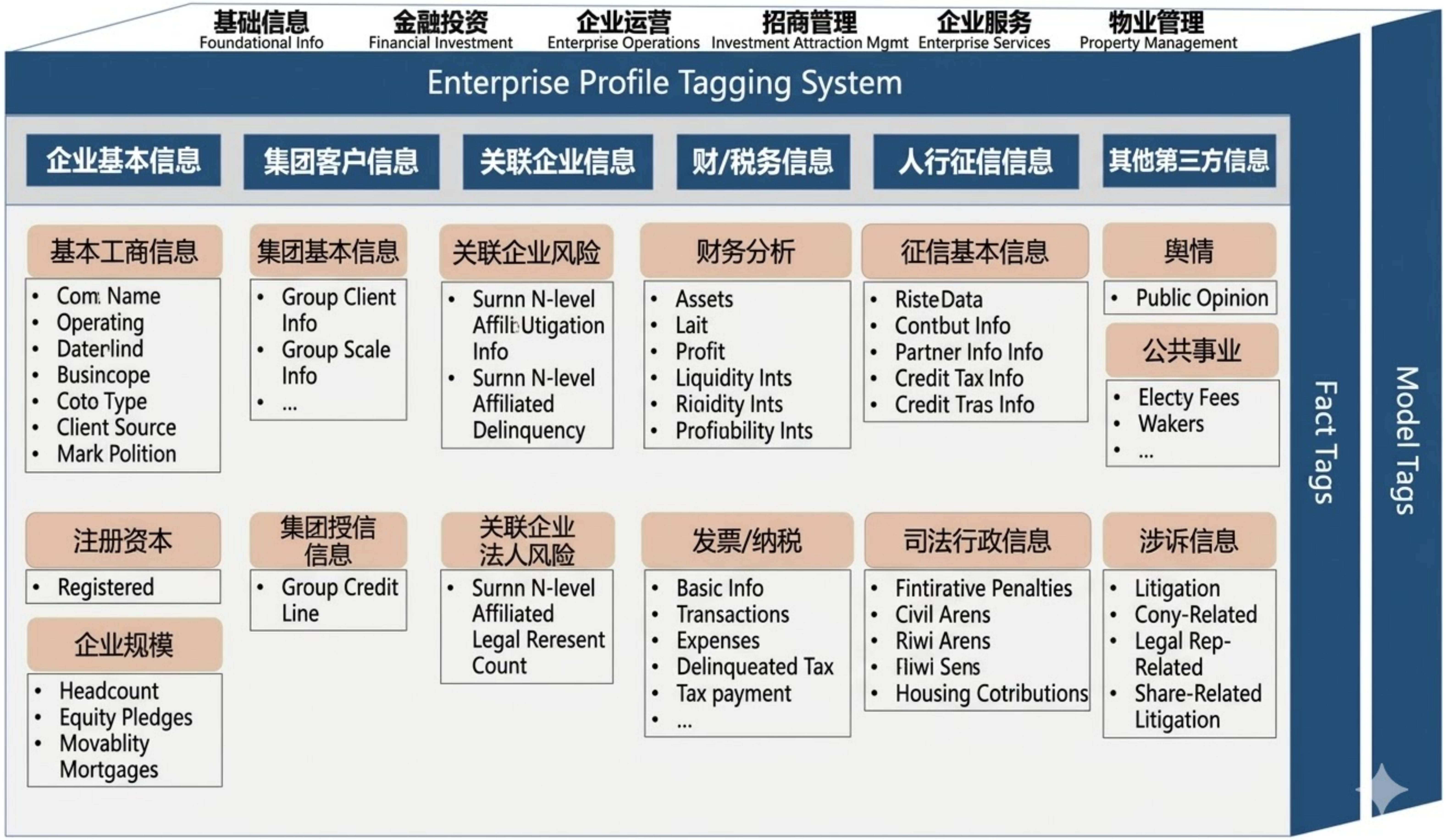
Continuing from the "People, Industry, and Space Operational Execution" framework established in the Business Opportunity Discovery phase, and incorporating data consolidation and pain point analysis, the following actions must be taken by **Dongsheng Bozhun** to effectively utilize and reconstruct authentic **Enterprise User Profiles**.



Design Thinking Workshop - Data Sorting

Digital Operational Decision-Making Platform
Enterprise User Profile System

Through the data streamlining conducted in the Design Thinking workshops, it has been decided to construct **Enterprise User Profiles** based on the following themes. Various departments will be able to share these profile dashboards according to their respective access permissions, allowing them to extract necessary insights for strategy formulation and the promotion of operational services.

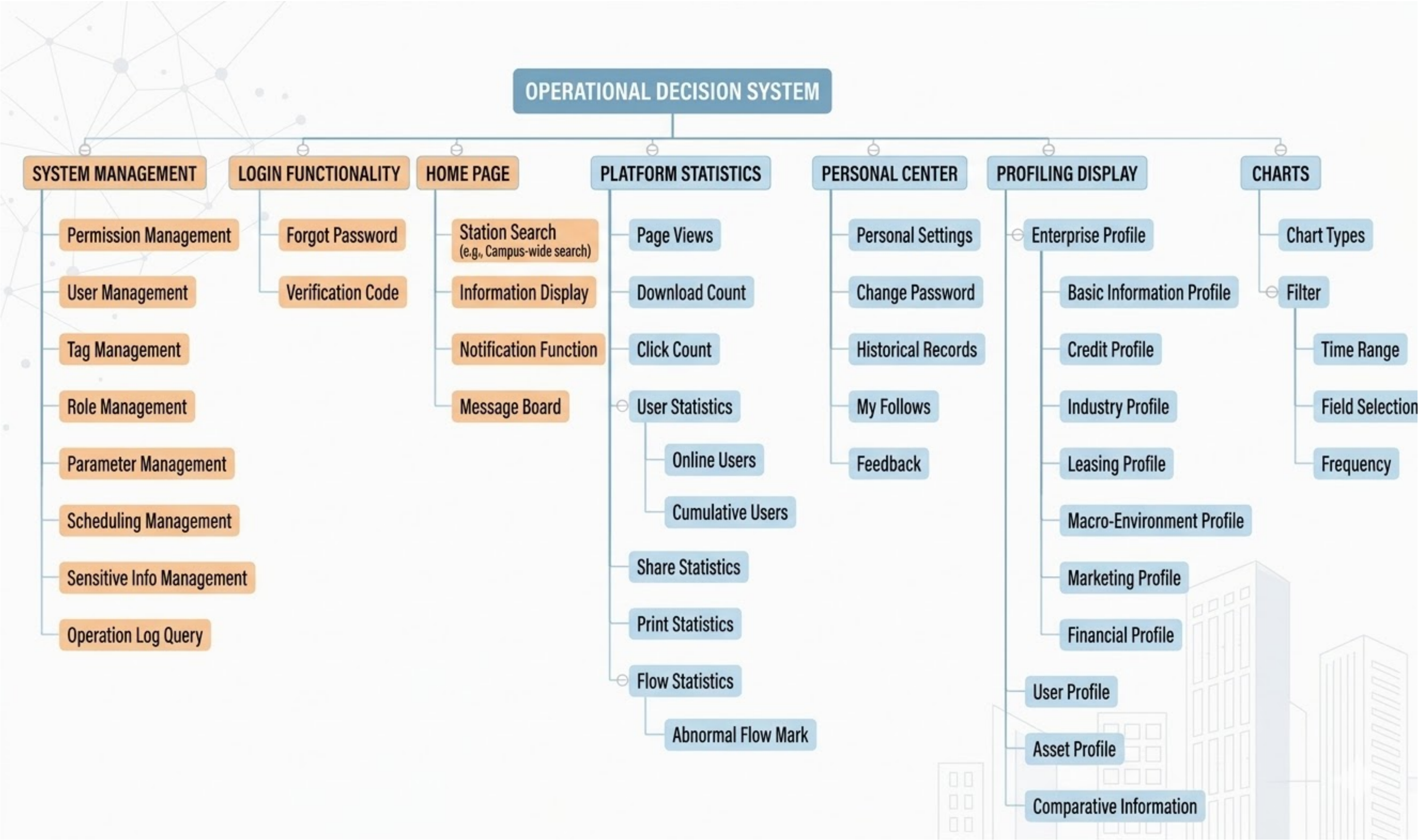


Enterprise User Profile System Map

Digital Operational Decision-Making Platform

Digital Operational Decision-Making Platform - Functional Structure

Beyond standard features such as **Secure Login, System Management, and Global Search**, the platform includes **Customizable Tagging, Enterprise User Profiling, and User Permission Management**. These functions allow users to define the specific dimensions for enterprise profile visualization and control the level of sensitive information disclosure. This ensures both complete operational flexibility and rigorous data confidentiality.



Functional Structure Map

Digital Operational Decision-Making Platform

Digital Operational Decision-Making Platform UI

Following the personalized configuration of the platform interface, the system will display various dashboards and datasets tailored to the user’s specific **Access Permissions**. Users can gain a clear understanding of all **Key Performance Indicators (KPIs)**, enabling them to formulate strategies more rapidly and maintain a deep insight into the evolving dynamics of the park.